

Week 4 - Part 1 - Osprey1st College Algebra - Summer 2024

Section covered: 3.4, 3.5 and 3.6

Definition of a Quadratic Equation A quadratic equation in x is an equation that can be written in the **general form**

$$ax^2 + bx + c = 0$$

where a, b , and c are real numbers, with $a \neq 0$. A quadratic equation in x is also called a **second-degree polynomial equation** in x .

The Zero-Product Principle

If the product of two algebraic expressions is zero, then at least one of the factors is equal to zero.

$$\text{If } AB = 0, \text{ then } A = 0 \text{ or } B = 0.$$

The Square Root Principle

$$\text{If } X^2 = k, \text{ then } X = \pm\sqrt{k}$$

Quadratic Formula

The solutions of a quadratic equation in general form $ax^2 + bx + c = 0$, are given by the **quadratic formula**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Types of Quadratic Equations, Most Efficient Solution Method.

1. The quadratic equation has no constant term ($c = 0$), $ax^2 + bx = 0$.

Most Efficient Solution Method: Factor and use the zero-product principle (you can always factor out at least x).

Problem 1. Solve each equation.

(a) $x^2 - 3x = 0$

$a=1$
 $b=-3$
 $c=0$

$$x(x-3)=0$$

$$x=0 \quad x-3=0$$

$$x=3$$

(b) $3x^2 = -12x$

$+12x \quad +12x$
 $3x^2 + 12x = 0$
 $3x(x+4) = 0$
 $3x=0 \quad x+4=0$
 $x=0 \quad x=-4$

2. The quadratic equation has no x term ($b = 0$). $ax^2 + c = 0$.

Most Efficient Solution Method: Solve for x^2 , or $(ax + b)^2$, and apply the square root property.

Problem 2. Solve each equation.

(a) $5x^2 + 1 = 51$

$x^2 = 49$
 $(7)^2 = 49$
 $(-7)^2 = 49$

$$\frac{5x^2}{5} = \frac{50}{5}$$

$$x^2 = 10$$

$$x = \pm \sqrt{10}$$

$$x = -\sqrt{10}, \sqrt{10}$$

(b) $(3x + 2)^2 + 9 = 0$

$-9 \quad -9$
 $\sqrt{(3x+2)^2} = \sqrt{-9}$
 $3x+2 = \pm 3i$
 $\frac{3x}{3} = \frac{-2 \pm 3i}{3}$
 $x = -\frac{2}{3} \pm \frac{3}{3}i = \boxed{-\frac{2}{3} \pm i} = -\frac{2}{3} + i, -\frac{2}{3} - i$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3. The quadratic equation has all three terms. $ax^2 + bx + c = 0$.

Most Efficient Solution Method: Check first if the equation can be factored easily. If not, apply quadratic formula.

Problem 3. Solve each equation.

(a) $4x^2 - 6x + 1 = 0$

$a = 4$
 $b = -6$
 $c = 1$

~~4×1
 $+4$
 -6~~

$$\frac{6 \pm \sqrt{6^2 - 4(4)(1)}}{2(4)}$$

$$\frac{6 \pm 2\sqrt{5}}{8}$$

$$x = \frac{3 \pm \sqrt{5}}{4}$$

$$\sqrt{36 - 16}$$

$$\sqrt{20}$$

$$\sqrt{4 \cdot 5}$$

$$2\sqrt{5}$$

(b) $x^2 = -11x - 10$

$+11x + 10$

$x^2 + 11x + 10 = 0$

$(x + 10)(x + 1) = 0$

$x + 10 = 0$

$x + 1 = 0$

$x = -10 \quad x = -1$

~~$+10$
 $+1$
 $+11$~~

(c) $x^2 - 6x = -10$

Problem 4. Solve each equation by the method of your choice.

(a) $(2v - 3)(5 - v) = 0$

(b) $3x^2 - 27 = 0$

(c) $4y^2 + 24y = 0$

(d) $(x - 6)(x + 2) = -18$

(e) * $u^2 - 4u + 4 = 0$

(f) $3x^2 - 12x + 12 = 0$

(g) * $3u^2 + 7u = -2$

$a=3$
 $b=7$
 $c=2$

$3u^2 + 7u + 2 = 0$

$(3u+6)(3u+1) = 0$

$(u+2)(3u+1) = 0$

$\begin{array}{c} 3 \times 2 \\ +6 \\ +6 \\ +7 \end{array}$

$u+2=0$

$3u+1=0$
 $3u=-1$

$u=-2 \quad u=-\frac{1}{3}$

(h) * $2w^2 + 19w + 44 = (w+4)^2$

(i) $\ast (v^2 + 4v + 3)(5v + 15) = 0$

(j) $\ast (v + 7)^2 - 24 = 0$

(k) $\ast 6x^2 + 8x - 1 = 0$

Discriminant

In the quadratic formula, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ the radicand $b^2 - 4ac$ is called the **discriminant** because, by knowing its value, we can discriminate among the possible number and type of solutions of a quadratic equation.

Discriminant $b^2 - 4ac$	Number and Type of Solutions
Positive	Two real solutions
Zero	One real solution
Negative	Two complex but not real solutions

Problem 5. Use the discriminant to determine the number and type of solutions of each quadratic equation. What is the number of real solutions?

(a) $x^2 + 2x + 1 = 0$

(b) $3x^2 - 1 = 0$

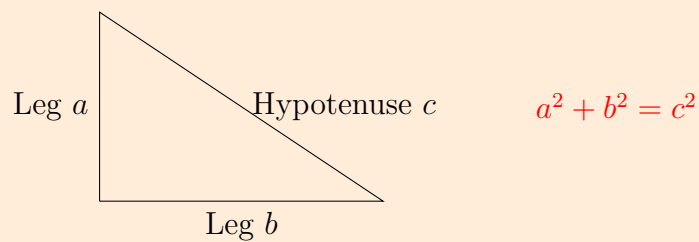
(c) $* -8x^2 + 8x - 3 = 0$

Problem 6. $*$ Write the quadratic equation whose roots are 2 and -4 , and whose leading coefficient is 3.

Problem 7. * The area of a rectangle is 45 ft^2 , and the length of the rectangle is 1 ft more than double the width. Find the dimensions of the rectangle.

Pythagorean Theorem

In a right triangle, the sum of the squares of the lengths of the legs is equal to the square of the length of the hypotenuse.



Problem 7. * The shorter leg of a right triangle is 7 m shorter than the longer leg. The hypotenuse is 7 m longer than the longer leg. Find the side lengths of the triangle.

Problem 8. * A ball is thrown from an initial height of 2 meters with an initial upward velocity of 7 m/s . The ball's height h (in meters) after t seconds is given by the following.

$$h = 2 + 7t - 5t^2$$

Find all values of t for which the ball's height is 3 meters. Round your answer(s) to the nearest hundredth.

Problem 9. * Find all excluded values for the expression. That is, find all values of u for which the expression is undefined. If there is more than one value, separate them with commas.

$$\frac{u^2 - 6u - 27}{u^2 - 9}$$